

CURRICULUM VITAE

Dr. Mahadev Mallikarjun Biradar

1. Personal Details

Name	Dr. Mahadev Mallikarjun Biradar
Present Designation	Associate Professor, Department of Mathematics
Institution	Basaveshwar Engineering College (Autonomous), Bagalkote
Mobile	9035246580
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ORCID ID	0000-0002-9897-8677
Google Scholar	PQDhCuoAAAAJ

2. Academic Qualifications

Degree	Specialization	University / Institution	Year
Ph.D.	Fluid Dynamics	Gulbarga University, Kalaburagi	2008
M.Phil.	Fluid Dynamics	Gulbarga University, Kalaburagi	2003
M.Sc.	Fluid Dynamics & CAGD	Gulbarga University, Kalaburagi	2002

3. Employment History

Designation and Institution	Period	Duration
Associate Professor, Basaveshwar Engineering College, Bagalkote	23-03-2009 to Present	Ongoing
Assistant Professor, Guru Nanak Dev Engineering College, Bidar	01-09-2005 to 21-03-2009	Approx. 3.5 years
Lecturer, Government Tool Room & Training Centre, Kalaburagi	April 2002 to August 2005	Approx. 3.5 years
Total Teaching / Academic Experience: Over 24 years		

4. Summary of Research and Academic Contributions

- **Research Publications:** 26
- **Books:** 2
- **Patents:** 4
- **Ph.D.:** Fluid Dynamics
- **Research Area:** Fluid Dynamics and related areas of Magnetohydrodynamics, nanofluids, heat and mass transfer, and mathematical modelling.

- **Administrative Leadership:** Head, Department of Mathematics (2015–2018)

5. Administrative and Academic Responsibilities Held

- **Head of the Department** — Mathematics, BEC (31-08-2015 to 10-12-2018).
- **Chairman, Board of Studies (BOS)** — Mathematics Department (31-08-2015 to 10-12-2018).
- **Program Officer** — Youth Red Cross, BEC Wing (Since 29-08-2012).
- **Member** — Board of Examiners (BOE) and Board of Studies (BOS).
- **Member** — Academic Council (31-08-2015 to 10-12-2018).
- **Coordinator** — QEEE Mathematics Department.
- **Member** — Malpractice Cases Consideration Committee.
- **Member** — Squad Committee.
- **Member** — Equivalence Committee.
- **Valuation Coordinator** — Examination Section.
- **Coordinator** — AICTE 3600 Feedback, Mathematics Department.
- **Coordinator** — College Website, Mathematics Department.

6. External Academic Appointments – VTU

- Member, VTU Ph.D. Research Advisory Committee (RAC).
- Member, VTU Board of Examiners (BOE) – Mathematics.
- Vice-Chancellor's Nominee, VTU Board of Studies (BOS).

DETAILED ACADEMIC PORTFOLIO

7. Research Publications

7.1 International Journal / Research Papers

1. **Mahadev Biradar**, A. B. Patil, M. Patil, V. S. Patil, R. B. Naik, and P. P. Humane (2026). “Perturbation analysis of MHD Prandtl fluid flow with viscous dissipation, thermal radiation and double diffusion.” *International Journal of Modelling and Simulation*, 1–15. DOI: [10.1080/02286203.2026.2708091](https://doi.org/10.1080/02286203.2026.2708091).
2. **Mahadev Biradar**, C. N. Chidananda, G. S. Roopa, J. K. Madhukesh, and G. K. Ramesh (2026). “Optimizing the heat performance of Casson ternary nanoliquid with radiative-magnetic-chemical reactive effects using Taguchi technology.” *Thermal Advances*, 7, 100155. ISSN: 3050-4635. DOI: [10.1016/j.thradv.2026.100155](https://doi.org/10.1016/j.thradv.2026.100155).
3. **Mahadev Biradar**, Amar B. Patil, Vishwambhar S. Patil, and Govind R. Rajput (2025). “Exploring anomalous diffusion in ternary nanofluids: a Maxwell model incorporating viscous dissipation and Nield boundary bioconvection.” *European Physical Journal Plus*, Springer, 140, 170, 1–19. DOI: [10.1140/epjp/s13360-025-02500-1](https://doi.org/10.1140/epjp/s13360-025-02500-1).
4. **Mahadev Biradar**, Jagadish V. Tawade, B. Haewon, M. Kaviyarasu, V. Govindan, et al. (2024). “Momentum, heat and mass transfer in the hydrodynamic electrically conducting fluid flow over a stretching sheet.” *Modern Physics Letters B*, World Scientific, 38(16), 2341011. DOI: [10.1142/S021798492341011X](https://doi.org/10.1142/S021798492341011X).
5. **Mahadev Biradar**, P. Priyanka, S. A. M. Abdelmohsen, Jagadish V. Tawade, A. M. M. Ashraf, R. Kumar, et al. (2023). “Multiple slip effects of MHD boundary-layer motion of a Casson nanoliquid over a penetrable linearly stretching sheet embedded in non-Darcian porous medium.” *International Journal of Modern Physics B*, World Scientific, 37(03), 2350022. DOI: [10.1142/S0217979223500227](https://doi.org/10.1142/S0217979223500227).
6. **Mahadev Biradar**, Jagadish V. Tawade, M. M. Biradar, and D. N. Manjunatha B. (2023). “Analysis of volume fraction of Casson nanofluid flow over a flat moving plate with thermal radiation and nonuniform heat source/sink.” *Journal of International Academy of Physical Sciences*, 27(3), 259–274.
7. **Mahadev Biradar**, Jagadish V. Tawade, and Rashmi Yala (2022). “MHD Casson nanofluid with nonuniform heat source/sink and viscous dissipation.” *Proceedings of the Jangjeon Mathematical Society*, 25(4), 387–400.
8. **Mahadev Biradar**, Padmini Kaji, and Jagadish V. Tawade (2022). “Numerical Solution of Hydrodynamic Electrically Conducting Fluid Flow over a Stretching Sheet.” *Mathematical Statistician and Engineering Applications*, 71(4), 12714–12730.
9. **Mahadev Biradar**, S. P. Pallavi, D. S. Sharanappa, V. Mb, and Jagadish V. Tawade (2022).

- “Heat generation and absorption effects on thin film flow of an unsteady Casson fluid across a penetrable flat plate.” *Mathematical Statistician and Engineering Applications*, 71(4), 13024–13041.
10. **Mahadev Biradar**, B. Manvi, Jagadish Tawade, S. Noeiaghdam, U. Fernandez-Gamiz, et al. (2022). “The effects of MHD radiating and non-uniform heat source/sink with heating on the momentum and heat transfer of Eyring-Powell fluid over a stretching.” *Results in Engineering*, Elsevier, 14, 100435, 1–11. DOI: [10.1016/j.rineng.2022.100435](https://doi.org/10.1016/j.rineng.2022.100435).
 11. **Mahadev Biradar**, S. S. Benal, Jagadish V. Tawade, and H. L. Allasi (2022). “Effects of the Magnetohydrodynamic Flow within the Boundary Layer of a Jeffery Fluid in a Porous Medium over a Shrinking/Stretching Sheet.” *Journal information to be verified*.
 12. **Mahadev Biradar**, Jagadish V. Tawade, and S. S. Benal (2021). “Casson Fluid Flow of Thin Liquid Film Beyond.” *Recent Trends in Mathematical Modeling and High Performance Computing*. Publication details to be verified.
 13. **Mahadev Biradar** and Patil Mallikarjun (2021). “Mixed Convective Flow With Variable Viscosity And Variable Thermal Conductivity In A Channel In The Presence Of First Order Chemical Reaction With Heat Generation Or Absorption.” *Journal of Applied Mathematics and Computational Mechanics*, 20(1), 91–102.
 14. **Mahadev Biradar**, Jagadish V. Tawade, and S. S. Benal (2021). “Influence of radiant heat and non-uniform heat source on MHD Casson fluid flow of thin liquid film beyond a stretching sheet.” *Recent Trends in Mathematical Modeling and High Performance Computing*. Publication details to be verified.
 15. **Mahadev Biradar**, S. B. Kerur, Jagadish V. Tawade, and B. K. (2020). “Effect of Non-uniform heat source/sink for the Casson nanofluid in presence of viscous dissipation over the porous stretching sheet.” *Journal of Engineering Sciences*. Volume and page details to be verified.
 16. **Mahadev Biradar**, Jagadish V. Tawade, and S. B. Kerur (2020). “Radiation effects on heat transfer of MHD Casson nanofluid flow over a porous stretching sheet in presence of non-uniform heat source/sink.” *Journal of Information and Computational Science*.
 17. **Mahadev Biradar**, Jagadish V. Tawade, and D. N. (2020). “Non-uniform heat source/sink and thermal radiation effects on liquid film flow of Casson fluid thin film over a stretching sheet.” *Journal of Engineering Sciences*.
 18. **Mahadev Biradar**, Jagadish V. Tawade, and S. S. Benal (2020). “A new approach of MHD flows for UCM fluid over a permeable moving plate with space and temperature dependent internal heat generation/absorption.” *AIP Conference Proceedings*, AIP Publishing, 2236, 1.
 19. **Mahadev Biradar** and Gousia Begum (2017). “Transient MHD Nanofluid Flow with

Chemical Reaction and Radiation Effects under Slip Boundary Conditions.” *International Journal of Enhanced Research in Science, Technology & Engineering*, 6(8), August 2017. ISSN: 2319-7463.

20. **Mahadev Biradar** (2015). “Flow and Heat Transfer of Micropolar and Viscous Fluid with Source or Sink in a Vertical Channel.” *International Journal on Emerging Technologies*, 6(2), 253.
21. **Mahadev Biradar** (2015). “Two fluid electromagneto convective flow and heat transfer between vertical wavy wall and a parallel flat wall.” *International Journal of Advanced Research in Engineering and Technology*.
22. **Mahadev Biradar** (2014). “Flow and heat transfer of micropolar and viscous fluid in a vertical channel.” *International Journal of Research in Engineering and Technology*, 3(3), 771–776.
23. **Mahadev Biradar** (2014). “Magneto-convective flow and heat transfer of two immiscible fluids between vertical wavy wall and a parallel flat wall.” *International Journal of Research in Engineering and Technology*, 3(1), 777–790.
24. **Mahadev Biradar**, J. Tawade, and M. M. Biradar (2013). “Two Fluid Convective Flow And Heat Transfer Between Vertical Wavy Wall And A Parallel Flat Wall.” *IJCAE*, 4(1), 144–163.
25. **Mahadev Biradar** (2013). “Convective Flow And Heat Transfer Of Composite Porous Medium Between Vertical Wavy Wall And A Parallel Flat Wall.” *IJCAE*, 4(1), 118–133.
26. **Mahadev Biradar**, J. C. Umavathi, and J. Prathap Kumar (2005). “Two Fluid Magnetoconvection in a Vertical Enclosure.” *Modelling, Measurement and Control B*, 74(3–4), 1–18.

8. Books Authored / Co-authored

1. **Mahadev Biradar** and Gousia Begum (2024). *Differential Equations: Methods & Applications*. Shineeks Publishers, pp. 1–189. ISBN: 979-8-89496-001-2.
2. Poonia, S., Begum, R., and **Mahadev Biradar** (2023). *Application of Integral Transforms in Mathematical Physics and Engineering*. Shineeks Publishers. Page details to be verified.

9. Patents

1. **AI-Powered Intelligent Financial Management System for Autonomous Cost Application**. Patent No. 202141036930, 2021. Status: Published.
2. **Vehicle Dynamics Analysis and Optimization of Electric All-Terrain Vehicle**. Patent No. 202111038638, 2021. Status: Published.

3. **Smart Stick Solution for Assisting Blind People Using Internet of Things.** Patent No. 202121042381, 2021. Status: Published.
4. **An Intelligent Attendance System with Geolocation and Anomaly Detection.** Patent No. 202521023180 A. Publication Date: 28/03/2025. Status: Published.

10. Declaration

I hereby declare that the information provided in this Curriculum Vitae and Academic Portfolio is true and correct to the best of my knowledge and belief. I understand that the information and supporting documents may be verified by the competent authority.

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